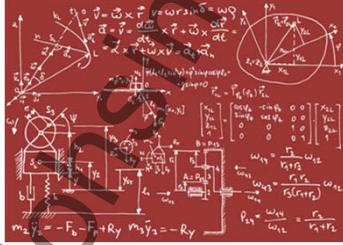


Vector Mechanics

- SCALAR & VECTOR
- VECTOR ALGEBRA
- VECTOR TYPES
- TAKE-HOME MESSAGE



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Scalar & Vector

SCALAR

What is a scalar?

A scalar is any positive or negative physical quantity that can be completely specified by its magnitude.

Examples?

length, mass, time, current

How do we represent a scalar?

10 meter, 50 Kilograms, 5 seconds,



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VECTORS

What is a vector?

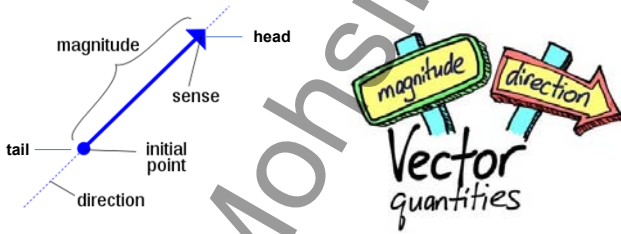
A vector is any physical quantity that requires both a magnitude and a direction for its complete description

Examples?

force, position, velocity, current density

How do we represent vectors?

Graphically



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SCALAR & VECTORS

What happens when we multiply a vector with a scalar?

Magnitude increases/decreases scalar times

Negative scalar reverses direction

Can a scalar be added to a vector?

No

How a vector is added to another vector

Triangle rule or "Head to Tail"

Parallelogram law of addition

Cartesian Vector Notation

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Vector Algebra

MATHEMATICAL FORMULATION

How do we represent vectors mathematically?

Magnitude & Direction

Let's call it vector $\vec{A}$

magnitude

head

sense

tail

initial point

direction

45°

Magnitude = $|\vec{A}| = 6$

Direction = $\angle \vec{A} = 45^\circ$

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Vector Algebra

EXAMPLE I

Suppose a student was at NU-FAST. He drives 50Km East and then he drives 10Km North followed by a 20km transit toward west.

I want to know, how far is he from campus?

$|P_1| = 50Km$
 $\angle P_1 = 0$

$|P_2| = 10Km$
 $\angle P_2 = 90^\circ$

$|P_3| = 20Km$
 $\angle P_3 = 180^\circ$

$|P_r| = ?$
 $\angle P_r = ?$

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Can I solve it?

Yes! Using Head & Tail Rule + Trigonometry I can?

$|\vec{P_r}| = \sqrt{30^2 + 10^2} = 31.623 Km$ $\angle \vec{P_r} = \tan^{-1}\left(\frac{10}{30}\right) = 18.635^\circ$

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Vector Algebra

EXAMPLE I

What if we were solving the following scenario?

$|P_r| = ?$
 $\angle P_r = ?$

$|P_1| = 50\text{Km}$
 $\angle P_1 = 0$

$|P_2| = 10\text{Km}$
 $\angle P_2 = 18^\circ$

$|P_3| = 53\text{Km}$
 $\angle P_3 = 161.6^\circ$

NOT so interesting for analytical work :(

What is the main difficulty?

Arbitrary directions

How can we solve this difficulty?

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Vector Algebra

SOLUTION

Vectors Addition/Subtraction is similar to Scalar Addition/Subtraction, if they are in the same direction

↓

Resolve vectors on to standard directions

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Vector Algebra

CARTESIAN VECTOR NOTATION

How else do we represent vectors mathematically?

Cartesian Vector Notation

Let's call it vector $\vec{A}$

tail head

magnitude

direction

$|A| = 6$

$\angle A = 45^\circ$

$|A_x| = 6 \cos(45^\circ)$

$|A_y| = 6 \sin(45^\circ)$

$$\vec{A} = |A_x| \hat{i} + |A_y| \hat{j} + |A_z| \hat{k}$$
$$\vec{A} = 6 \cos(45^\circ) \hat{i} + 6 \sin(45^\circ) \hat{j} + 0 \hat{k}$$

$\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors in the direction of x,y and z-axis respectively

$$\vec{A} = \begin{bmatrix} |A_x| \\ |A_y| \\ |A_z| \end{bmatrix} \quad \vec{A} = \begin{bmatrix} 6 \cos(45^\circ) \\ 6 \sin(45^\circ) \\ 0 \end{bmatrix} = \begin{bmatrix} 4.243 \\ 4.243 \\ 0 \end{bmatrix}$$

Preferred writing style!

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EXAMPLE I CONT.

Suppose a student was at NU-FAST. He drives 50Km East and then he drives 10Km North followed by a 20km transit toward west.

I want to know, how far is he from campus?

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$|P_1| = 50 \text{ km}$

$|P_2| = 10 \text{ km}$

$|P_3| = 20 \text{ km}$

$|P_r| = ?$

$\angle P_r = ?$

$|P_{rx}| = 30 \text{ km}$

$|P_{rz}| = 10 \text{ km}$

$$\vec{P}_r = \vec{P}_1 + \vec{P}_2 + \vec{P}_3$$
$$\vec{P}_r = \begin{bmatrix} 50 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 10 \\ 0 \end{bmatrix} + \begin{bmatrix} -20 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 30 \\ 10 \\ 0 \end{bmatrix} \text{ km}$$

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Vector Algebra

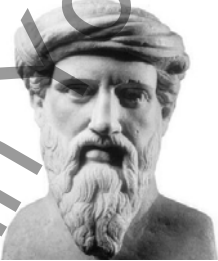
MAGNITUDE & DIRECTION

Given rectangular components, how can we find the actual vector?

Pythagorean theorem

$$|\vec{A}| = \sqrt{|\vec{A}_x|^2 + |\vec{A}_y|^2}$$
$$\angle \vec{A} = \tan^{-1} \left(\frac{|\vec{A}_y|}{|\vec{A}_x|} \right)$$

$$|\vec{A}| = \sqrt{|\vec{A}_x|^2 + |\vec{A}_y|^2 + |\vec{A}_z|^2}$$
$$\cos \alpha = \frac{|\vec{A}_x|}{|\vec{A}|} \quad \cos \beta = \frac{|\vec{A}_y|}{|\vec{A}|} \quad \cos \gamma = \frac{|\vec{A}_z|}{|\vec{A}|}$$



Pythagoras of Samos
(570 -495 BC)

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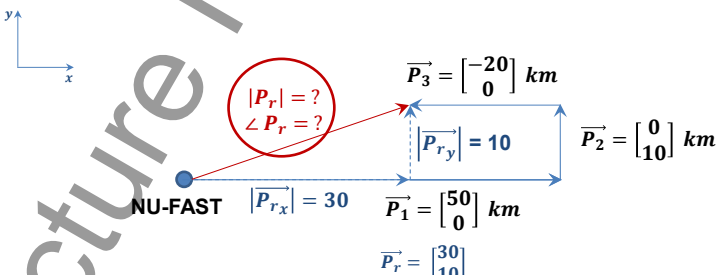
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Vector Algebra

EXAMPLE I CONT.

Suppose a student was at NU-FAST. He drives 50Km East and then he drives 10Km North followed by a 20km transit toward west.
I want to know, how far is he from campus?



$\vec{P}_1 = \begin{bmatrix} 50 \\ 0 \end{bmatrix} \text{ km}$ $\vec{P}_2 = \begin{bmatrix} 0 \\ 10 \end{bmatrix} \text{ km}$ $\vec{P}_3 = \begin{bmatrix} -20 \\ 0 \end{bmatrix} \text{ km}$ $\vec{P}_r = \begin{bmatrix} 30 \\ 10 \end{bmatrix}$ $|\vec{P}_r| = \sqrt{|\vec{P}_{rx}|^2 + |\vec{P}_{ry}|^2} = \sqrt{30^2 + 10^2} = 31.623 \text{ Km}$ $\angle \vec{P}_r = \tan^{-1} \left(\frac{|\vec{P}_{ry}|}{|\vec{P}_{rx}|} \right) = \tan^{-1} \left(\frac{10}{30} \right) = 18.635^\circ$

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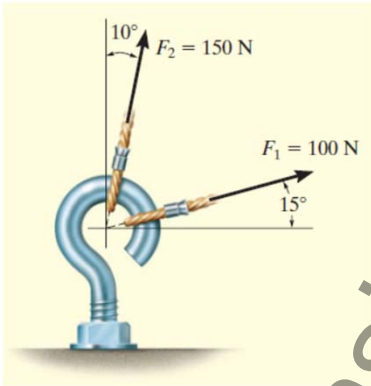
2.2 Vector Operations

2.3 Vector Addition of Forces



Vector Algebra

EXAMPLE II



Determine the magnitude and direction of the resultant force.

Parallelogram Law

Cartesian Vector Notation

[1] - Russell C. Hibbeler, Engineering mechanics: Statics and dynamics 12th edition, Example 2-1

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[illegible]

Vector Algebra

2.4 Addition of a System of Coplanar Forces

Determine the x and y component of forces F_1 and F_2 acting on the boom and express each force as a Cartesian vector.

Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, Example 2.5

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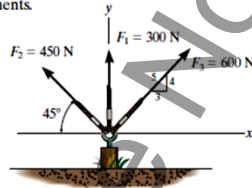
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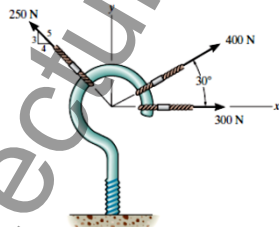
Vector Algebra

2.4 Addition of a System of Coplanar Forces

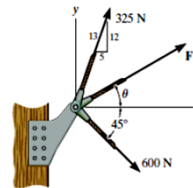
F2-7. Resolve each force acting on the post into its x and y components.



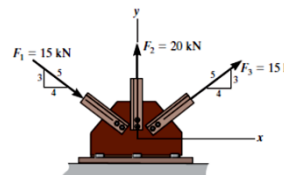
F2–8. Determine the magnitude and direction of the resultant force.



F2-10. If the resultant force acting on the bracket is to be 750 N directed along the positive x axis, determine the magnitude of F and its direction θ .



F2-12. Determine the magnitude of the resultant force and its direction θ measured counterclockwise from the positive x axis.



Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pg 38

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Vector Algebra

2.4 Addition of a System of Coplanar Forces

PRACTICE PROBLEMS

*2–39. Determine the magnitude of F_1 and its direction θ so that the resultant force is directed vertically upward and has a magnitude of 800 N.

*2–40. Determine the magnitude and direction measured counterclockwise from the positive x axis of the resultant force of the three forces acting on the ring A . Take $F_1 = 500\text{ N}$ and $\theta = 20^\circ$.

2–51. If $F_1 = 150\text{ N}$ and $\phi = 30^\circ$, determine the magnitude of the resultant force acting on the bracket and its direction measured clockwise from the positive x axis.

*2–52. If the magnitude of the resultant force acting on the bracket is to be 450 N directed along the positive y axis, determine the magnitude of F_1 and its direction ϕ .

*2–53. If the resultant force acting on the bracket is required to be a minimum, determine the magnitudes of F_1 and the resultant force. Set $\phi = 30^\circ$.

Russell C. Hibbeler, Engineering mechanics: Statics and dynamics 12th edition, pg 39-42

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Vector Algebra

UNIT VECTOR

What is a Unit Vector?

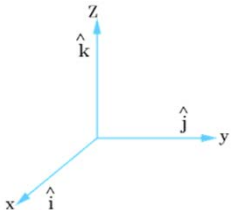
A vector of Unit Magnitude

How to find a Unit Vector along a given vector $\vec{A}$?

Dividing the given vector by its magnitude

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$

A unit vector is often used to represent directions



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Vector Algebra

2.8 Force Vector Directed Along a Line

EXAMPLE IV

The man shown in the figure pulls on
the cord with a force of 70N. Represent
this force acting on the support A as a
Cartesian vector [1]

$$\overrightarrow{AB} = \begin{Bmatrix} 3 \\ -2 \\ -6 \end{Bmatrix} m$$

$$\hat{\overline{AB}} = \frac{\overline{AB}}{|\overline{AB}|} = \begin{Bmatrix} 3 \\ -2 \\ -6 \end{Bmatrix} \times \frac{1}{7} = \begin{Bmatrix} 0.429 \\ -0.286 \\ -0.857 \end{Bmatrix}$$

$$\vec{F} = 70 \times \begin{Bmatrix} 0.429 \\ -0.286 \\ -0.857 \end{Bmatrix} = \begin{Bmatrix} 30 \\ -20 \\ -60 \end{Bmatrix} N$$

[1] - Russell C. Hibbeler, Engineering mechanics: Statics and dynamics 11th edition SI, Example 2.13, pp. 51

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Vector Algebra

2.8 Force Vector Directed Along a Line

PRACTICE FUNDAMENTAL PROBLEMS

F2-19. Express the position vector $\mathbf{r}_{AB}$ in Cartesian vector form, then determine its magnitude and coordinate direction angles.

F2-21. Express the force as a Cartesian vector.

F2-22. Express the force as a Cartesian vector.

F2-23. Determine the magnitude of the resultant force at A.

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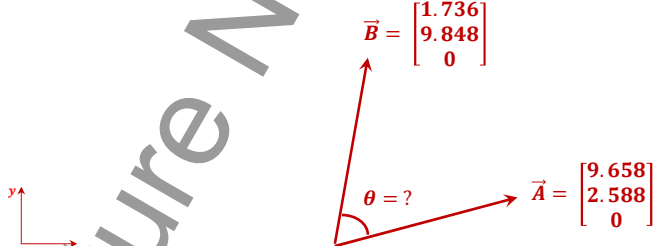
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Vector Algebra

VECTOR PRODUCTS

Given two vectors, how can one find the angle between them?



$$\vec{B} = \begin{bmatrix} 1.736 \\ 9.848 \\ 0 \end{bmatrix}$$

$$\vec{A} = \begin{bmatrix} 9.658 \\ 2.588 \\ 0 \end{bmatrix}$$

$\theta = ?$

By finding angle of F_1 and F_2 because it's a 2D problem

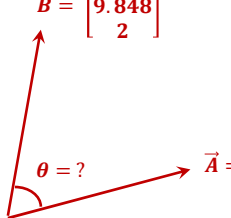
$\theta = 65^\circ$

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Vector Algebra

VECTOR PRODUCTS

Given two vectors, how can one find the angle between them?



$$\vec{B} = \begin{bmatrix} 1.736 \\ 9.848 \\ 2 \end{bmatrix}$$

$$\vec{A} = \begin{bmatrix} 9.658 \\ 2.588 \\ 5 \end{bmatrix}$$

$\theta = ?$

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos(\theta)$$

$\vec{A} \cdot \vec{B} = 52.253$	$ \vec{A} = 10.198$	$ \vec{B} = 11.179$
$\theta = 62.72^\circ$		

Vector Algebra

2.9 Dot Product:

How to find components of a vector parallel and perpendicular to a line?

$$\vec{B} = \begin{bmatrix} 1.736 \\ 9.848 \\ 2 \end{bmatrix}$$

0.863
0.231
0.447

Dot Product :)

How?

$$\vec{B} \cdot \hat{A} = |\vec{B}| |\hat{A}| \cos(\theta) = |\vec{B}| \cos(\theta)$$

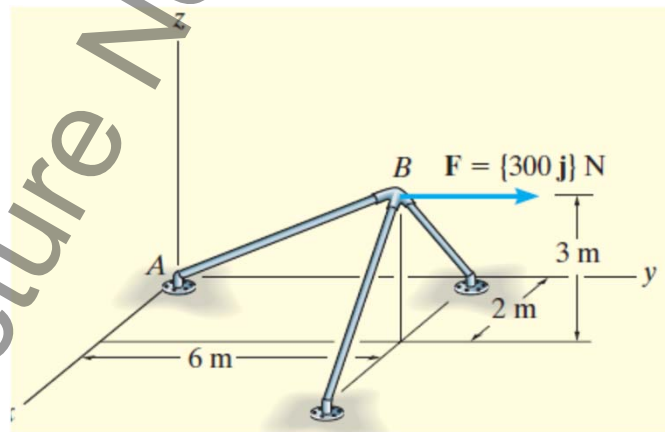
Magnitude of Component of $\vec{B}$ along $\hat{A} = \vec{B} \cdot \hat{A} = 4.666$

Vector Algebra

2.9 Dot Product:

EXAMPLE V

Determine the components of the force F parallel and perpendicular to member AB [1]



[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 11th edition SI, Example 2.16, pp. 61

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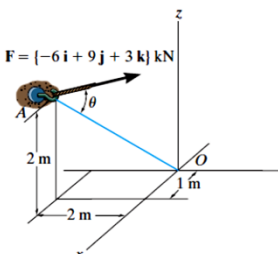
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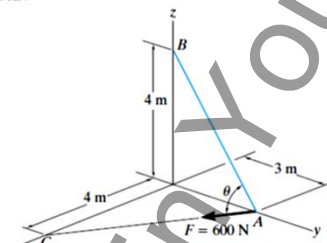
2.9 Dot Product

PRACTICE FUNDAMENTAL PROBLEMS

F2-25. Determine the angle θ between the force and the line AO .

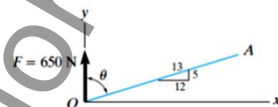


F2-26. Determine the angle θ between the force and the line AB .



F2-27. Determine the angle θ between the force and the line OA .

F2-28. Determine the component of projection of the force along the line OA .



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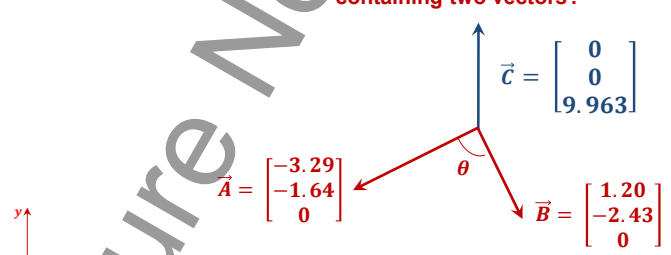
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Vector Algebra

VECTOR PRODUCTS

How to find a vector, which is perpendicular to a plane containing two vectors?



$\vec{C} = \begin{bmatrix} 0 \\ 0 \\ 9.963 \end{bmatrix}$

$\vec{A} = \begin{bmatrix} -3.29 \\ -1.64 \\ 0 \end{bmatrix}$

$\vec{B} = \begin{bmatrix} 1.20 \\ -2.43 \\ 0 \end{bmatrix}$

Cross Product :

$$\vec{C} = \vec{A} \times \vec{B}$$

$$\vec{C} = |\vec{A}||\vec{B}|\sin(\theta)\hat{u}_C$$

$$\vec{C} = \begin{bmatrix} 0 \\ 0 \\ 9.963 \end{bmatrix}$$

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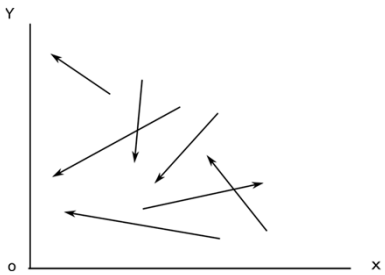
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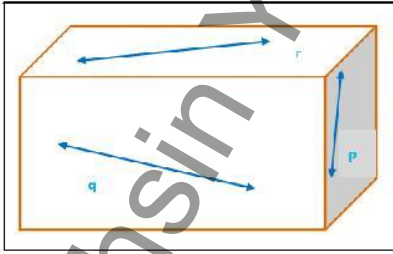
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COPLANAR

Coplanar vectors in the same plane



Coplanar



NOT Coplanar

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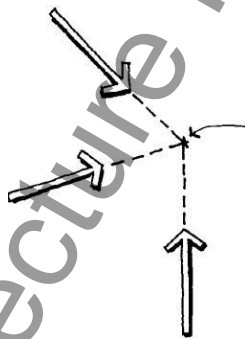
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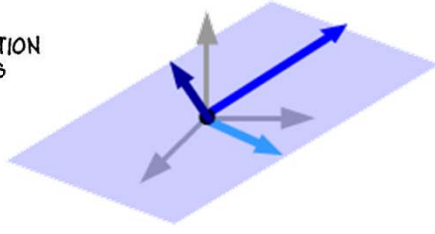
Vector Types

CONCURRENT

Concurrent vector system is one in which the lines of action of all the vectors intersect at a common point O



LINES OF ACTION MEET AT THIS POINT.



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PROBLEM SOLVING STEPS

- 1 – Mathematical Formulation
- 2 – Solution Strategy
- 3 – Actual Solution
- 4 – Justification & Reflection

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CARTESIAN VECTOR NOTATION

Vectors Addition/Subtraction is similar to Scalar Addition/Subtraction, if they are in the same direction

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APPLICATION OF DOT PRODUCT

- 1 - Finding the angle between two given vectors
- 2 – Finding the projection of one vector on to another

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Lecture Notes by Mohsin Yousuf